

KARTHIK M DAIVADNYA

AI/ML Engineer | Software Engineer

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PROFESSIONAL SUMMARY

B.Tech Computer Science & Engineering graduate with hands-on experience in data analysis, statistics, and machine learning, including data cleaning, exploratory data analysis (EDA), feature engineering, and predictive model development using Python (Pandas, NumPy, Scikit-learn) and SQL. Skilled at translating structured and semi-structured data into actionable insights and interactive dashboards (Power BI, Matplotlib), with working knowledge of data quality, validation, and benchmarking practices. Strong foundation in statistics and probability, data structures & algorithms, and REST API design, plus additional exposure to Deep Learning and Generative AI (LangChain, RAG). Comfortable leveraging AI-assisted development tools (GitHub Copilot, Claude) to accelerate experimentation and productivity. Collaborative and curious, with a track record of Agile cross-functional teamwork and clear documentation of analysis, methodologies, and results.

EDUCATION

Bapuji Institute of Engineering and Technology (BIET), Davangere

B.Tech in Computer Science & Engineering — CGPA: 8.76 / 10.0

2022 – 2026

EXPERIENCE

AI and Data Science Internship — Boston Institute of Analytics

Feb – Jun 2026

- Analyzed and explored structured and semi-structured datasets to identify patterns, trends, and data quality issues, performing data preprocessing, cleaning, transformation, and feature engineering in Python (Pandas, NumPy) following clean, maintainable coding practices.
- Developed and evaluated statistical and machine learning models to support business decision-making, and built Power BI and Matplotlib visualizations to communicate predictive analytics insights to stakeholders.
- Wrote Python scripts and SQL queries for data validation and quality checks, helping define benchmarks and validation criteria to ensure accurate, reliable analysis on structured datasets.
- Collaborated with a remote, cross-functional team to translate raw data into business reports and interactive dashboards, participating in reviews, iterative debugging, and clear documentation of findings and methodologies.
- Leveraged AI-assisted development tools to accelerate experimentation and streamline repetitive data-analysis tasks, improving turnaround time on deliverables.

PROJECTS

Multi-Agent AI System — Autonomous Task Orchestration Platform

2026

Tech Stack: Python, LangGraph, LangChain, Groq API, Streamlit, Git, GitHub

- Designed and implemented a multi-agent orchestration system in Python using LangGraph and LangChain, applying OOP principles to structure agent roles, state, and communication.
- Integrated Groq's free-tier LLM API to power reasoning agents, building modular, testable components and debugging inter-agent workflows end-to-end.
- Built a Streamlit front end for real-time interaction and deployed the codebase to GitHub with version-controlled commits, documentation, and a reproducible project structure.

DocMind — AI-Powered Document Assistant (RAG + NLP)

2026

Tech Stack: Python, LangChain, Mistral AI, FAISS, FastAPI, React.js, REST APIs

- Built a Retrieval-Augmented Generation (RAG) pipeline with LangChain and the Mistral AI LLM, transforming unstructured PDFs into a queryable knowledge base for natural-language Q&A.
- Engineered the NLP retrieval layer with FAISS vector embeddings, tuning chunking and retrieval parameters through iterative experimentation to optimize response accuracy.
- Developed and deployed a FastAPI backend exposing REST APIs, paired with a React.js dashboard (session analytics, PDF export) for technical and non-technical users.

AeroSense — Predictive Maintenance ML Platform

2026

Tech Stack: Python, Pandas, NumPy, Scikit-learn, FastAPI, React.js, REST APIs

- Built and deployed a predictive maintenance platform on the NASA CMAPSS turbofan dataset (20,000+ sensor records), with a FastAPI backend and React.js frontend serving real-time predictions via REST APIs.
- Preprocessed multivariate sensor data, performed statistical feature engineering, and trained 4 ML models (Random Forest, Gradient Boosting, XGBoost, Linear Regression) using Scikit-learn.
- Optimized model performance through comparative evaluation, achieving a best R^2 of 0.975, and documented the full analysis workflow and methodology in Jupyter Notebook for maintainability.

SalesPulse — Sales Analytics & Data Pipeline

2025

Tech Stack: Python, Pandas, SQL, SQLite, Power BI, Git

- Designed an end-to-end ETL pipeline in Python (Pandas) to clean, validate, and consolidate raw multi-source sales data (Excel/CSV) into a structured SQLite database.
- Wrote and validated SQL queries to aggregate revenue, regional performance, and customer trends, powering an interactive Power BI dashboard with KPI cards and drill-down filters.
- Automated weekly/monthly report generation with a Python script, reducing manual reporting effort and ensuring consistent, on-time delivery of data-driven insights.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL

Data Science & ML: Pandas, NumPy, Scikit-learn, Statistics & Probability, Data Analysis & Visualization, Feature Engineering, Data Quality & Validation, Machine Learning, Deep Learning Fundamentals, NLP, Generative AI (LLMs, RAG)

Visualization & BI: Power BI, Matplotlib, Jupyter Notebook

Frameworks & Libraries: FastAPI, Flask, React.js, Node.js, LangChain

Databases: SQL, MySQL, SQLite, Relational Databases, MongoDB (NoSQL)

Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, REST API Design

Tools & Platforms: Git, GitHub, Postman, VS Code, PySpark, Agile/Scrum collaboration

AI Tools: LangChain, FAISS, GitHub Copilot, Claude, Mistral AI

CERTIFICATIONS

- AI Agent Architect — Learner to Builder, IBM (2025)
- Machine Learning Basics — Infosys Springboard (2025)
- DevOps Foundation — Infosys Springboard (2025)
- MongoDB Basics (2024)
- Networking Basics — Cisco (2024)